Renormalization of 3D Luttinger action

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Three-dimensional Luttinger fermions have quadratic band touching dispersion with Lifshitz-like dynamical scaling $z = 2$. We extend the single-particle Luttinger Hamiltonian to include the effects of minimally coupled gauge fluctuations and illustrate the long-wavelength behavior of the combined interacting system. Within the parameter space of our model, we discover an analog of the non-Fermi liquid phase found in Luttinger semimetals with static Coulomb interaction [1]. We further show that dynamical gauge fluctuations can also lead to non-relativistic Lifshitz scaling for the emergent photons in our theory. With an RG analysis, we describe the stability and scaling properties of its fixed points in the presence of additional cubic-symmetry allowed perturbations.

I. THE ACTION AND ITS CONSERVED CURRENTS

A. The fermion sector

Let us consider an U(1) gauge theory coupled to fermions with symmetry protected quadratic band touching. The action without the gauge fields has been discussed before [1, 2]. In the rotation and particle-hole symmetric limit, using $N_\Gamma = (d-1)(d/2+1)$ Clifford matrices $\{\Gamma^a\}$ of dimensions 4×4, the Euclidean Lagrangian is expressed as [3, 4]:

$$\begin{aligned}
\mathcal{L}_{\rm L} &= \psi^\dagger \left[\partial_0 + \frac{d_a(-i\partial)}{2m}\Gamma^a\right]\psi, \\
d_a\left(-i\partial\right) &= -\sqrt{\frac{d}{2(d-1)}}\Lambda_a^{ij}\partial_i\partial_j,\ \{\Gamma^a,\Gamma^b\} = 2\delta_{ab},
\end{aligned} \tag{1}$$

where $\{\Lambda_a\}$ are the $(d-1)(d+2)/2$ real, symmetric $d \times d$ Gell-Mann matrices [3, 4]. In three dimensional momentum space the band dispersions are [4],

$$\begin{aligned}
d_1(k) &= \frac{3}{2}\left(k_x^2 - k_y^2\right),\ d_2(k) = \sqrt{3}k_xk_y, \\
d_3(k) &= \sqrt{3}k_xk_z,\ d_4(k) = \sqrt{3}k_yk_z, \\
d_5(k) &= \frac{1}{2}\left(2k_z^2 - k_x^2 - k_y^2\right)
\end{aligned} \tag{2}$$

In our notation the greek indices span the space-time and the roman indices only span the space. It can be shown that $\Gamma^{ab} = \frac{1}{2}[\Gamma^a,\Gamma^b]$ are the $N_\Gamma(N_\Gamma-1)/2$ generators of $SO(N_\Gamma)$ rotations [5]. Certain useful identities of the Γ matrices are as follows:

$$\begin{aligned}
\Gamma^b\Gamma^a\Gamma^b &= (2-N_\Gamma)\Gamma^a \\
\Gamma^c\Gamma^a\Gamma^b\Gamma^c &= (N_\Gamma-4)\Gamma^a\Gamma^b + 4\delta_{ab} \\
\Gamma^d\Gamma^a\Gamma^b\Gamma^c\Gamma^d &= -2\Gamma^c\Gamma^b\Gamma^a - (N_\Gamma-4)\Gamma^a\Gamma^b\Gamma^c
\end{aligned} \tag{3}$$

The symmetric Gell-mann matrices obey the identities [2–4],

$$\Lambda_a^{ij}\Lambda_a^{kl} = (\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}) - \frac{2}{d}\delta_{ij}\delta_{kl},$$

$$\mathrm{tr}\left[\Lambda_a \Lambda_b\right] = 2\delta_{ab}, \tag{4}$$

Using these identities it can be shown in a simple way that [4]

$$\begin{aligned} d_a(p)d_a(q) &= \frac{d}{d-1}\left(\left(\vec{p}_\perp \cdot \vec{q}_\perp\right)^2 - \frac{1}{d}p_\perp^2 q_\perp^2\right), \\ \implies d_a(q)d_a(q) &= q_\perp^4. \end{aligned} \tag{5}$$

The global U(1) symmetry of the action (1) is manifest with the global transformations $\psi \mapsto e^{ie\alpha}\psi$. The corresponding global currents are given by,

$$\begin{aligned} j_0 &= ie\psi^\dagger\psi, \\ j_i &= ie\sqrt{\frac{d}{2(d-1)}}\frac{\Lambda_a^{ij}}{2m}\left[\psi^\dagger\Gamma^a\partial_j\psi - \partial_j\psi^\dagger\Gamma^a\psi\right], \end{aligned} \tag{6}$$

with the current conservation condition,

$$\partial_\mu j_\mu = 0, \tag{7}$$

dictated by the symmetry. We seek to elevate the global U(1) symmetry of the model to a local U(1) gauge symmetry by introducing vector gauge fields A_μ which are minimally coupled with the action. The inclusion of the minimal coupling is achieved by replacing all the normal derivatives of the action with covariant derivatives $D_\mu = \partial_\mu + ieA_\mu$. With the covariant derivatives, the action reads as:

$$\mathcal{L}'_{\mathrm{L}} = \psi^\dagger\left[\partial_0 + ieA_0 - \sqrt{\frac{d}{2(d-1)}}\Lambda_a^{ij}\Gamma^a\frac{D_iD_j}{2m}\right]\psi, \tag{8}$$

with the rule that under the local U(1) transformations both the fermionic fields and the vector gauge fields transform simultaneously with the formula,

$$\mathrm{U}(1) : \psi \mapsto e^{ie\varphi(x)}\psi, \tag{9}$$

$$\mathrm{U}(1) : A_\mu(x) \mapsto A_\mu(x) - \partial_\mu\varphi(x), \tag{10}$$

where $x = (\tau, \vec{x})$ denotes the space-time coordinate. Similarly for the frequency-momentum space we shall use the notation $k = (k_0, \vec{k})$. It is straightforward to check that under the gauge transformations [Eq. (10)] the action of the covariant derivative changes as,

$$\mathrm{U}(1) : D_\mu\psi \mapsto (\partial_\mu + ieA_\mu(x) - ie\partial_\mu\varphi(x))e^{ie\varphi(x)}\psi = e^{ie\varphi(x)}D_\mu\psi, \tag{11}$$

which in turn ensures that the minimally coupled action written down above (8) remains invariant under the gauge transformation.

B. The photon sector

With the fermionic sector well defined we need a gauge invariant description of the gauge field action. We are interested in an U(1) gauge theory in (3+1) dimension. The gauge invariant Euclidean Maxwell action is given by [5, 6]

$$\mathcal{L}_{\text{Maxwell}} = \frac{1}{4}F_{\mu\nu}^2 + j^\mu A_\mu, \tag{12}$$

with the matter current j^μ defined above (6). However, we are in a situation where the Lorentz invariance is absent in the fermionic sector, and naturally, space and time has different engineering dimensions. This scenario has been explored in the context of the pure U(1) gauge theory of pyrochlore compounds [7, 8]. From the fermion sector it is imperative that before any loop correction the engineering dimension of space and time is respectively $[x^{-1}] = 1$ and $[\tau^{-1}] = z_0 = 2$. The appropriate U(1) gauge theory in this situation goes by the name of $z_0 = 2$ Lifshitz electrodynamics [9, 10] and its properties have been studied in details [9–12]. The Lagrangian is most compactly written in Euclidean space-time as,

$$\mathcal{L}_{\text{U(1)}} = \frac{1}{2}F_{0i}F_{0i} + \frac{1}{4}F_{ij}\left(u^2 + v^2\left[-\nabla^2\right]\right)F_{ij} + j_\mu A_\mu, \text{ with } F_{\mu\nu} = D_\mu A_\nu - D_\nu A_\mu. \tag{13}$$

As a consequence of gauge invariance, the photon propagator obtained from this action is singular. To determine the photon propagator we need to work on a fixed gauge. In a Lorentz invariant theory a gauge fixing function $G(A) = \partial_\mu A_\mu$ can be set to zero because under a gauge transformation $A_\mu(x) \mapsto A_\mu^\alpha(x) = A_\mu(x) + \partial_\mu\alpha(x)$ the function transforms as

$$G(A_\mu^\alpha(x)) = \partial_\mu A_\mu(x) + \partial^2\alpha(x), \tag{14}$$

and we can always choose and $\alpha(x)$ such that the function vanishes. The gauge fixing function can in principle be completely arbitrary. In the current case a suitable gauge fixing function for the action is given by [10] -

$$G(A_\mu, \omega) = \partial_0 A_0(x) + L_g\partial_i A_i(x) - \omega(x) = 0, \tag{15}$$

where $\omega(x)$ is an arbitrary function and we have defined a differential operator $L_g = u^2 + v^2\left[-\nabla^2\right]$, In the path integral we must only sum over the unique gauge configurations dictated by the gauge orbit $\omega(x)$ (15). Various different gauge field configurations which are related to each other through the $\alpha(x)$ reparameterization must be counted once. The original functional integral is given by

$$Z_{\text{U(1)}}[j] = \int \mathcal{D}A e^{-\int_x \mathcal{L}_{\text{U(1)}}(A_\mu, j_\mu)}, \tag{16}$$

where $\int_x = \int dx_0 d^d x_\perp$ is a short-hand notation for the $d+1$ dimensional space-time integral. We wish to integrate over gauge field configurations which satisfy the gauge fixing condition $G(A_\mu, \omega) = 0$. We shall therefore sum over all gauge field reparameterization $A_\mu(x) \mapsto A_\mu^\alpha(x)$ that leaves the functional integral invariant but also obey the gauge fixing condition. This can be carried out by the Fadeev-Popov prescription where one uses the identity [13],

$$1 = \int \mathcal{D}\alpha \det\left(\frac{\delta G(A_\mu^\alpha, \omega)}{\delta\alpha}\right)\delta\left(G(A_\mu^\alpha, \omega)\right), \tag{17}$$

to modify the partition function of the gauge fields with a uniform unity weight,

$$Z_{\mathrm{U}(1)}[j] = \int \mathcal{D}\alpha\mathcal{D}A e^{-\int_x \mathcal{L}_{\mathrm{U}(1)}(A_\mu, j_\mu)} \det\left(\frac{\delta G(A^\alpha_\mu,\omega)}{\delta\alpha}\right)\delta\left(G(A^\alpha_\mu,\omega)\right). \tag{18}$$

With $\frac{\delta G(A^\alpha_\mu,\omega)}{\delta\alpha} = \partial_0^2 + L_g\partial_i^2$ the determinant in this expression factors out. With the intention to make the photon propagator more simple, a non-local gauge orbit can be chosen for our Lorentz-violating gauge field action [10–12] with a gauge fixing function,

$$G(A_\mu,\varphi) = \partial_0 A_0(x) + L_g\partial_i A_i(x) - L_g^{1/2}\varphi(x), \tag{19}$$

where we shall sum over all functions $\varphi(x)$ with a Gaussian weight centered around $\varphi(x) = 0$. Finally, we make a gauge transformation $A_\mu \mapsto A_\mu + \partial_\mu\alpha$ in the functional integral over the gauge fields to obtain,

$$\begin{aligned} Z_{\mathrm{U}(1)}[j] &= \int \mathcal{D}\varphi e^{-\int_x \frac{\varphi(x)^2}{2\xi}} \int \mathcal{D}\alpha \det\left(\partial_0^2 + L_g\partial_i^2\right) \\ &\quad\times \int \mathcal{D}A^\alpha e^{-\int_x \mathcal{L}_{\mathrm{U}(1)}(A^\alpha_\mu, j_\mu)}\delta\left(\partial_0 A^\alpha_0(x) + L_g\partial_i A^\alpha_i(x) - L_g^{1/2}\varphi(x)\right) \\ &= \det\left(\partial_0^2 + L_g\partial_i^2\right)\int \mathcal{D}\alpha \int \mathcal{D}A^\alpha e^{-\int_x \mathcal{L}_{\mathrm{U}(1)}(A^\alpha_\mu, j_\mu)} e^{-\int_x \frac{\left(L_g^{-1/2}\partial_0 A^\alpha_0(x) + L_g^{1/2}\partial_i A^\alpha_i(x)\right)^2}{2\xi}}. \end{aligned} \tag{20}$$

It is clear in the second line that $\alpha(x)$ has become a dummy parameter and naturally the functional integral over $\alpha(x)$ simply factors out. The procedure therefore achieves its goal of summing over unique gauge field configurations. The Feynman-'t Hooft gauge $\xi = 1$ further simplifies the gauge fixed action,

$$\mathcal{L}_{\mathrm{gf}} = \frac{1}{2}F_{0i}F_{0i} + \frac{1}{4}F_{ij}L_gF_{ij} + \frac{1}{2}\left(L_g^{-1/2}\partial_0 A_0 + L_g^{1/2}\partial_i A_i\right)^2, \tag{21}$$

and yields a simple non-singular gauge field propagator to [11],

$$D^{\mathrm{gf}}_{\mu\nu}(k) = \langle A_\mu(-k)A_\nu(k)\rangle = \left(\delta_{\mu0}\delta_{\nu0}\frac{L_g(k_\perp^2)}{k_0^2 + k_\perp^2 L_g(k_\perp^2)} + \delta_{\mu i}\delta_{\nu j}\frac{\delta_{ij}}{k_0^2 + k_\perp^2 L_g(k_\perp^2)}\right), \tag{22}$$

where $k_\perp^2 = k_i^2$ is the length of the momentum $\vec{k}_\perp$. Henceforth, we shall use the gauge-fixed propagator D^{gf} without referring to the gauge fixing directly.

II. RG ANALYSIS

A. The action

The renormalization group analysis of the fully interacting problem begins by separating the action in the free and the interacting vertices part. For this analysis we must consider a gauge-invariant Lagrangian with symmetry allowed terms that are closed under renormalization. For this reason, in addition to the terms discussed so far we also include a Pauli coupling term,

$$\frac{\kappa}{2}\psi^\dagger\Gamma^{ab}\psi\eta^i_{ab}\epsilon_{ijk}F_{jk}, \tag{23}$$

with $\Gamma^{ab} = \frac{1}{2i}\left[\Gamma^a, \Gamma^b\right]$ and $\eta^i_{ab} = -\eta^i_{ba}$ are the symbols that relate Spin 3/2 SU(2) matrices with Γ_{ab} [2],

$$S^i = \eta^i_{ab}\Gamma^{ab}, \left[\eta^i, \eta^j\right] = -\frac{1}{4}\epsilon^{ij}_k \eta^k.$$

We shall need some more properties of η^i matrices from their defining relation. For $D = 4$ dimensional Γ^{ab} matrices we have the formula $\mathrm{tr}\left[\Gamma^{ab}\Gamma^{cd}\right] = D\left(\delta_{ac}\delta_{bd} - \delta_{ad}\delta_{bc}\right)$, which we may use to find,

$$\begin{aligned} \mathrm{tr}\left[S^i S^j\right] &= -2D\eta^i_{ab}\eta^j_{ba}, \\ \mathrm{tr}\left[\eta^i \eta^j\right] &= -\frac{5}{2D}\delta^{ij} = -\frac{5}{8}\delta^{ij}. \end{aligned} \tag{24}$$

It also follows that $\eta^i\eta^i = cI$, because this operator commutes with all the generators η^i. Using the trace identity we have,

$$\eta^i\eta^i = -\frac{15}{8N_\Gamma}I \tag{25}$$

To relate η^i_{ab} symbols with the Λ^{ij}_a matrices we need the formula [14],

$$\Gamma^a = \frac{1}{\sqrt{2d(d-2)}}\{S^i, S^j\}\Lambda^{ij}_a, \tag{26}$$

and use it in conjunction with the η and Λ matrices. We find,

$$\begin{aligned} \eta^i_{ab} &= \frac{1}{8}\mathrm{Tr}\left[S^i\Gamma^{ab}\right], \\ &= \frac{1}{32d(d-2)i}\Lambda^{mn}_a\Lambda^{rs}_b\mathrm{Tr}\left[S^i\left[\{S^m, S^n\}, \{S^r, S^s\}\right]\right] \end{aligned} \tag{27}$$

The S^i commutators in the last line can be extracted by inverting Eq. (26) above. The Pauli coupling κ is relevant in the infrared at $d = 3$. Together, we have the action,

$$\begin{aligned} \mathcal{L} =& \mathcal{L}_{\mathrm{gf}} + \mathcal{L}_{\mathrm{L}} + \mathcal{L}_{\mathrm{int}}, \\ \mathcal{L}_{B,\mathrm{g}} =& \frac{1}{2}F_{B,0i}F_{B,0i} + \frac{1}{4}F_{B,ij}\left(u_B^2 + v_B^2\left[-\nabla^2\right]\right)F_{B,ij}, \\ \mathcal{L}_{B,\mathrm{L}} =& \psi^\dagger_B\left[\partial_0 + \frac{\not{d}\left(-i\partial\right)}{2m_B}\right]\psi_B, \\ \mathcal{L}_{B,\mathrm{int}} =& ie_B\Big[\psi^\dagger_B A_{B,0}\psi_B - \sqrt{\frac{d}{2(d-1)}}\frac{1}{2m_B}\left(\psi^\dagger_B \not{\Lambda}^{ij} A_{B,i}\partial_j\psi_B - \left(\partial_j\psi^\dagger_B\right)\not{\Lambda}^{ij}A_{B,i}\psi_B\right)\Big] \\ &+ \frac{e_B^2}{2m_B}\psi^\dagger_B\not{d}\left(A\right)\psi_B + \frac{\kappa_B}{2}\psi^\dagger_B\Gamma^{ab}\psi_B\eta^i_{ab}\epsilon_{ijk}\left(\partial_j A_{B,k} - \partial_k A_{B,j}\right), \end{aligned} \tag{28}$$

where $u_B^2 > 0, v_B^2 > 0$ for a positive definite (bounded) bosonic action, and we have introduced the Dirac slash notation, $P_a\Gamma^a = \not{P}$. The zero temperature Euclidean theory is embedded in (d+1) dimensions. The tree level engineering dimension of space and time can be set at $[x] = -1$ and

$[\tau] = -z_0$. The engineering dimensions of the various fields and couplings of the theory are as follows,

$$\begin{aligned}
&[\psi] = d/2,\ \left[m^{-1}\right] = z_0 - 2,\ [A_0] = (d + z_0 - 2)/2,\ [A_i] = (d - z_0)/2,\\
&\left[e^2\right] = 2 + z_0 - d,\ \left[u^2\right] = 2z_0 - 2,\ \left[v^2\right] = 2z_0 - 4,\\
&[\kappa] = (3z_0 - d - 2)/2
\end{aligned} \tag{29}$$

With $z_0 = 2$ all the couplings of the action are dimensionless except from e^2 and u. It is clear that the theory is strongly coupled with the relevant coupling u in any spatial dimensions with $z_0 = 2$. On the other hand the choice of $z_0 = 1$ turns the fermion bands flat. To investigate whether the theory has Wilsoninan fixed point within its parameter space we manually tune the relevant coupling to a critical value $u_c = 0$ and study the effective theory within a critical hypersurface parameterized by the remaining marginal couplings. The critical gauge field action therefore becomes,

$$\mathcal{L}_{\rm g}^{c} = \frac{1}{2} F_{B,0i} F_{B,0i} + \frac{v_B^2}{4} F_{B,ij} \left[-\nabla^2\right] F_{B,ij} \tag{30}$$

However, the gauge charge e is also strongly relevant in the physical spatial dimension of interest $d = 3$. For perturbative control on the effective critical theory we follow Moon et al. [1] to work in $d = 4 - \epsilon$ dimensions such that all the couplings of the theory are marginally relevant or irrelevant except from u which we critically tune to zero. Along with the compact temporal integral (Euclidean time-cycle) we are therefore undertaking a $5 - \epsilon$ expansion of the theory. We start by renormalizing the bare fields and parameters as,

$$\begin{aligned}
&\psi_B = \sqrt{Z_\psi}\psi,\ A_{B,\mu} = \sqrt{Z_A} A_\mu,\\
&m_B^{-1} = Z_{m^{-1}} m^{-1},\ e_B \mapsto Z_e e,\ v_B \mapsto Z_v v,\\
&\kappa_B = Z_\kappa \kappa
\end{aligned} \tag{31}$$

The renormalized Lagrangian is therefore given by,

$$\begin{aligned}
\mathcal{L}_{\rm crit} &= \mathcal{L}_{\rm g}^{c} + \mathcal{L}_{\rm L} + \mathcal{L}_{\rm int},\\
\mathcal{L}_{\rm g}^{c} &= \frac{Z_A}{2} F_{0i} F_{0i} + \frac{Z_A Z_v v^2}{4} F_{ij} \left[-\nabla^2\right] F_{ij},\\
\mathcal{L}_{\rm L} &= Z_\psi \psi^\dagger \left[\partial_0 + Z_{m^{-1}} \frac{\not{d}(-i\partial)}{2m}\right] \psi,\\
\mathcal{L}_{\rm int} &= Z_e Z_\psi \sqrt{Z_A} i e \psi^\dagger \Big[A_0 - Z_{m^{-1}} \sqrt{\frac{d}{2(d-1)}} \frac{\not{\Lambda}^{ij} \left(\partial_i A_j + A_j \partial_i + A_i \partial_j\right)}{2m}\Big] \psi\\
&\quad + Z_e^2 Z_\psi Z_A Z_{m^{-1}} \frac{e^2}{2m} \psi^\dagger \not{d}(A)\, \psi + Z_\kappa Z_\psi \sqrt{Z}_A \frac{\kappa}{2} \psi^\dagger \Gamma^{ab} \psi \eta_{ab}^i \epsilon_{ijk} \left(\partial_j A_k - \partial_k A_j\right),
\end{aligned} \tag{32}$$

Gauge invariance imposes further constraints on the renromalization parameters. We have,

$$\begin{aligned}
&Z_\psi = Z_e Z_\psi \sqrt{Z_A} \text{ and } Z_\psi Z_{m^{-1}} = Z_e^2 Z_\psi Z_A Z_{m^{-1}},\\
\implies &Z_e \sqrt{Z}_A = 1.
\end{aligned} \tag{33}$$

We redefine the remaining independent renormalization parameters as,

$$Z_\psi = Z_1,$$

$$
\begin{aligned}
&Z_\psi Z_{m^{-1}} = Z_2,\\
&Z_A = Z_3,\\
&Z_A Z_v = Z_4,\\
&Z_\kappa Z_\psi \sqrt{Z}_A = Z_5.
\end{aligned}
\tag{34}
$$

As a consequence it is possible to find the one-loop renormalization group coefficients of the theory by studying its two-point functions alone. Furthermore we shall expand around the tree level values of these renormalization factors, $Z_i = 1 + \delta Z_i = 1 + \delta_i$. This splits the Lagrangian between its tree level form and counter terms:

$$
\begin{aligned}
\mathcal{L} =& \mathcal{L}^c_{\rm g} + \mathcal{L}_{\rm L} + \mathcal{L}_{\rm int} + \mathcal{L}_{\rm ct},\\
\mathcal{L}^c_{\rm g} =& \frac{1}{2} F_{0i} F_{0i} + \frac{v^2}{4} F_{ij} \left[-\nabla^2\right] F_{ij},\\
\mathcal{L}_{\rm L} =& \psi^\dagger \left[\partial_0 + \frac{\not{d}(-i\partial)}{2m}\right] \psi,\\
\mathcal{L}_{\rm int} =& ie\Big[\psi^\dagger A_0 \psi - \sqrt{\frac{d}{2(d-1)}} \frac{1}{2m} \left(\psi^\dagger \not{\Lambda}^{ij} A_i \partial_j \psi - \left(\partial_j \psi^\dagger\right) \not{\Lambda}^{ij} A_i \psi\right)\Big] + \frac{e^2}{2m} \psi^\dagger \not{d}(A) \psi\\
&+ \kappa \psi^\dagger \sigma_{ij} \left(\partial_i A_j - \partial_j A_i\right) \psi,\\
\mathcal{L}_{\rm ct} =& \delta_3 \frac{1}{2} F_{0i} F_{0i} + \delta_4 \frac{v^2}{2} F_{ij} F_{ij}\\
&+ \delta_1 \psi^\dagger \left[\partial_0 + ieA_0\right] \psi + \delta_2 \psi^\dagger \left[\frac{\not{d}(-i\partial + eA)}{2m}\right] \psi + \delta_5 \frac{\kappa}{2} \psi^\dagger \Gamma^{ab} \psi \eta^i_{ab} \epsilon_{ijk} \left(\partial_j A_k - \partial_k A_j\right)
\end{aligned}
\tag{35}
$$

We have all the pieces in hand to evaluate the renormalized perturbation series diagrammatically in the momentum space. We adopt the following convention for the fourier transformation of the fields,

$$
A_\mu(x) = \int \frac{d^{d+1}k}{(2\pi)^{d+1}} A_\mu(k) e^{ik.x}, \tag{36}
$$

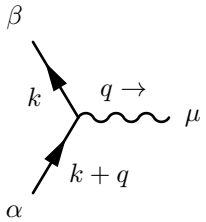
$$
\psi(x) = \int \frac{d^{d+1}k}{(2\pi)^{d+1}} \psi(k) e^{ik.x}. \tag{37}
$$

$$
\tag{38}
$$

As a consistent choice of notation, momentum entering a feynman diagram vertex shall be considered positive, while the momentum leaving a vertex shall be considered negative. Defining a function, $\not{d}^i(k) = \sqrt{\frac{d}{2(d-1)}} \not{\Lambda}^{ij} k_j$, we can compactly write down the momentum space ferynman rules for the action:

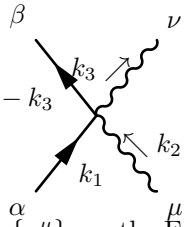
$$
\alpha \xrightarrow[k]{} \beta = \langle \psi_\alpha(k) \psi^\dagger_\beta(k) \rangle = G_{\alpha\beta}(k) = \left[\frac{-ik_0 + \not{d}(k)/(2m)}{k_0^2 + k_\perp^4/(4m^2)}\right]_{\alpha\beta}
$$

$$
\mu \overset{k \rightarrow}{\sim\sim\sim} \nu = \langle A_\mu(-k) A_\nu(k) \rangle = D^{\rm gf}_{\mu\nu}(k) = \left(\delta_{\mu 0}\delta_{\nu 0} \frac{v^2 k_\perp^2}{k_0^2 + v^2 k_\perp^4} + \delta_{\mu i}\delta_{\nu j} \frac{\delta_{ij}}{k_0^2 + v^2 k_\perp^4}\right),
$$



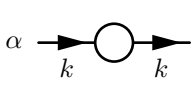
$$= C^{\mu}_{\alpha\beta}(k+q,-q,-k) = -e\Big[i\delta_{\mu 0} + \frac{\left(\not{d}^i(k+q) - \not{d}^i(-k)\right)}{2m}\delta_{\mu i}\Big]_{\alpha\beta}$$

$$- \kappa \left[\Gamma^{ab}\right]_{\alpha\beta} \eta^i_{ab}\epsilon_{ijk}\left(-iq_j\right)\delta_{\mu k},$$

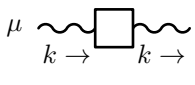


$$= V^{\mu\nu}_{\alpha\beta}(k_1,k_2,-k_3,-k_1-k_2+k_3) = -\frac{e^2}{m}\sqrt{\frac{d}{2(d-1)}}\delta_{\mu i}\delta_{\nu j}\left[\not{\Lambda}^{ij}\right]_{\alpha\beta}, \quad (39)$$

where $\{u^\mu\}$ are the Euclidean unit vectors and the factor of 2 in the quartic vertex comes from the symmetry permutation of the photon fields. The Feynman diagram of the counter term vertices are as follows:

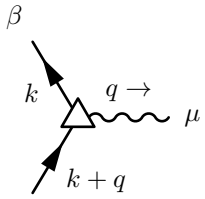


$$= \delta_\Sigma = -\left[\delta_1\left(ik_0\right) + \delta_2\left(\frac{\not{d}(k)}{2m}\right)\right]_{\alpha\beta},$$



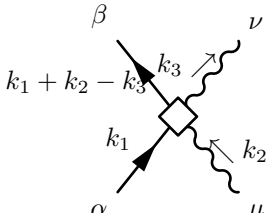
$$= \delta_\Pi = -\frac{\delta_3}{2}\left\{k^2\delta_{\mu\nu} - k_\mu k_\nu - \delta_{\mu i}\delta_{\nu j}\left(k^2_\perp\delta_{ij} - k_i k_j\right)\right\}$$

$$- \delta_4\delta_{\mu i}\delta_{\nu j}\frac{v^2k^2_\perp}{2}\left(\delta_{ij}k^2_\perp - k_i k_j\right),$$



$$= \delta_C = -e\Big[\delta_1\left(i\delta_{\mu 0}\right) + +\delta_2\left(\frac{\left(\not{d}^i(k+q) - \not{d}^i(-k)\right)}{2m}\right)\delta_{\mu i}\Big]_{\alpha\beta}$$

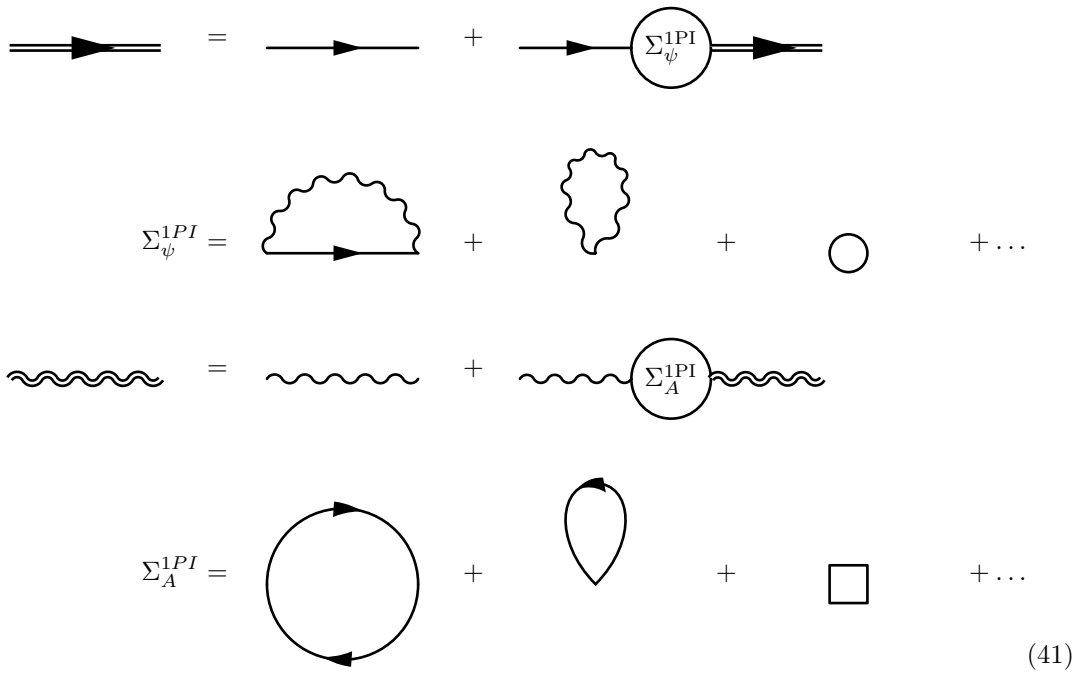
$$- \delta_5\kappa\left[\Gamma^{ab}\right]_{\alpha\beta}\eta^i_{ab}\epsilon_{ijk}\left(-iq_j\right)\delta_{\mu k},$$



$$= \delta_V = -\delta_2\frac{e^2}{m}\sqrt{\frac{d}{2(d-1)}}\delta_{\mu i}\delta_{\nu j}\left[\not{\Lambda}^{ij}\right]_{\alpha\beta} \quad (40)$$

It is clear that the counter terms are determined from the self-energy correction to the inverse

propagator alone. The full renormalized perturbation series can now be setup in terms of these diagrams. Following standard technique we sum up all one-particle irreducible (1PI) diagrams [13] and find,



(41)

In terms of the vertex functions and the propagators the diagrams are evaluated to

$$\begin{aligned}
\Sigma_\psi^{\rm 1L}(k) &= \int_q C^\mu(k+q,-q,-k)G(k+q)C^\nu(k,q,-k-q)D^{\rm gf}_{\mu\nu}(q) \\
&\quad + \int_q V^{\mu\nu}(k,q,-q,-k)D^{\rm gf}_{\mu\nu}(q) + \delta_\Sigma, \\
\Sigma_A^{\mu\nu,{\rm 1L}}(k) &= -\int_q {\rm tr}\left[C^\mu(q+k,-k,q)G(q+k)C^\nu(q,k,-q-k)G(q)\right] \\
&\quad - \int_q {\rm tr}\left[V^{\mu\nu}(q,k,-k,-q)G(q)\right] + \delta_\Pi,
\end{aligned} \tag{42}$$

where $\int_q = \int \frac{dq_0}{2\pi}\frac{d^d q_\perp}{(2\pi)^d}$ is the shorthand for the $d+1$ dimensional integral over internal frequency-momenta q. To determine the renormalization parameters we shall demand the finiteness of the tree level action with respect to an artificial renormalization scale μ. Which implies that the divergent pieces in the self energy expressions should identically cancel. This criteria shall determine the divergent counter terms of the action.

It can be shown that the only external momentum dependence of self-energies are through the set of functions $(k_0, d_a(k), d^i_a(k))$. Therefore, we may perform a Taylor expansion to obtain the leading order external momentum dependence of the self-energies in terms of this function. We can

introduce some additional parameters t, u, v and obtain,

$$\Sigma(tk_0, ud_a(k)) \tag{43}$$

B. Useful formulae for the one-loop integrals

We shall compute the frequency part of the loop integrals exactly as they involve no divergences. For the remaining d dimensional momentum integral we shall first resolve the complicated denominators of the frequency-integrated loop expressions using the Feynman's parametric integral formulae[15]:

$$\begin{aligned}
\frac{1}{A^a B^b} &= \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \int_0^1 dx \frac{x^{a-1}(1-x)^{b-1}}{[Ax + B(1-x)]^{a+b}}, \\
\frac{1}{A_1^{a_1} \dots A_n^{a_n}} &= \frac{\Gamma(a_1 + \dots + a_n)}{\Gamma(a_1) \dots \Gamma(a_n)} \int_0^1 dx_1 \cdots \int_0^1 dx_n \frac{\delta(1 - (x_1 + \dots + x_n)) x_1^{a_1 - 1} \dots x_n^{a_n - 1}}{(x_1 A_1 + \dots + x_n A_n)^{a_1 + \dots + a_n}}.
\end{aligned} \tag{44}$$

Following the Feynman parameterization, a momentum shift puts the generic loop integral in the form, $\int_{q_\perp} f_1(q_\perp^2) f_2(q_i, k_\mu)$, which can be further reduced by considering the integral identity,

$$\int \frac{d^d q}{(2\pi)^d} f(q^2) q_i q_j = \frac{\delta_{ij}}{d} \int \frac{d^d q}{(2\pi)^d} f(q^2) q^2. \tag{45}$$

In fact we shall need similar identities involving multiple products of internal momentum with open indices. In such types of integral the only non-zero contribution comes from cases where all the internal momentum indices are pairwise contracted. It is straightforward to show this with the help of the Laplace transform of the (well-behaved) function $f(q^2)$. With the transform the integrals of the type,

$$K_{ijkl} = \int \frac{d^d q}{(2\pi)^d} f(q^2) q_i q_j q_k q_l, \tag{46}$$

$$= \int_0^\infty dt \int \frac{d^d q}{(2\pi)^d} \tilde{f}(t) e^{-q^2 t} q_i q_j q_k q_l, \tag{47}$$

can be recast in terms of a generating variable Y_i such that,

$$\begin{aligned}
K_{ijkl} &= \frac{\partial^4}{\partial Y_i \partial Y_j \partial Y_k \partial Y_l} \left[\int_0^\infty dt \int \frac{d^d q}{(2\pi)^d} \tilde{f}(t) e^{-q^2 t + Y_i q_i} \right] \Bigg|_{Y_i \to 0}, \\
&= \frac{\partial^4}{\partial Y_i \partial Y_j \partial Y_k \partial Y_l} \left[\int_0^\infty dt \tilde{f}(t) \frac{e^{Y_i^2/(4t)}}{(2\pi)^{d/2} (2t)^{d/2}} \right] \Bigg|_{Y_i \to 0}, \\
&= \int_0^\infty dt \frac{\tilde{f}(t)}{(2\pi)^{d/2} (2t)^{d/2}} \left(\frac{\delta_{ij}}{2t} \frac{\delta_{kl}}{2t} + \frac{\delta_{ik}}{2t} \frac{\delta_{jl}}{2t} + \frac{\delta_{il}}{2t} \frac{\delta_{jk}}{2t} \right).
\end{aligned} \tag{48}$$

On the other hand we find that,

$$\int \frac{d^d q}{(2\pi)^d} f(q^2) q^4 = \int_0^\infty dt \int \frac{d^d q}{(2\pi)^d} \tilde{f}(t) e^{-q^2 t} q^4,$$

$$
\begin{aligned}
&= \int_0^\infty dt \tilde{f}(t) \frac{\partial^2}{\partial^2 t} \int \frac{d^d q}{(2\pi)^d} e^{-q^2 t}, \\
&= \int_0^\infty dt \tilde{f}(t) \frac{\partial^2}{\partial^2 t} \left(\frac{1}{(2\pi)^{d/2} (2t)^{d/2}} \right), \\
&= \int_0^\infty dt \frac{\tilde{f}(t)}{(2\pi)^{d/2} (2t)^{d/2}} \left(\frac{d(d+2)}{4t^2} \right).
\end{aligned}
\tag{49}
$$

Combining these two results we find that,

$$
K_{ijkl} = \int \frac{d^d q}{(2\pi)^d} f(q^2) q_i q_j q_k q_l = \frac{1}{d(d+2)} \int \frac{d^d q}{(2\pi)^d} f(q^2) q^4 \left(\delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk} \right), \tag{50}
$$

and similarly,

$$
K_{ijklmn} = \int \frac{d^d q}{(2\pi)^d} f(q^2) q_i q_j q_k q_l q_m q_n = \frac{1}{d(d+2)(d+4)} \int \frac{d^d q}{(2\pi)^d} f(q^2) q^4 \left(\delta_{ij}\delta_{kl}\delta_{mn} + \dots \right). \tag{51}
$$

C. The fermion self-energy

We start with the one-loop renormalized fermion self-energy $\Sigma_\psi^{\rm 1L}(k)$. The full expression is,

$$
\begin{aligned}
\Sigma_\psi^{\rm 1L}(k) = \int_q \Bigg[& e \left[i\delta_{\mu 0} + \frac{\not{d}^i(q+2k)}{2m} \delta_{\mu i} \right] \\
& \times \left[\frac{-i(k_0+q_0) + \not{d}(k+q)/(2m)}{(k_0+q_0)^2 + d(k+q)^2/(4m^2)} \right] \\
& \times e \left[i\delta_{\nu 0} + \frac{\not{d}^i(q+2k)}{2m} \delta_{\nu i} \right] \\
& \times \left(\delta_{\mu 0}\delta_{\nu 0} \frac{v^2 q_\perp^2}{q_0^2 + v^2 q_\perp^4} + \delta_{\mu i}\delta_{\nu j} \frac{\delta_{ij}}{q_0^2 + v^2 q_\perp^4} \right) \Bigg] - \left[\delta_1 (ik_0) + \delta_2 \left(\frac{\not{d}(k)}{2m} \right) \right].
\end{aligned}
\tag{52}
$$

The contribution from the quartic vertex identically vanishes in this case. We shall consider the case for non-zero external frequency and momenta separately. For the $k = (k_0, 0)$ we obtain,

$$
\begin{aligned}
{\rm tr}\left[\Sigma_\psi^{\rm 1L}(k_0, 0) \right] = & \int_q {\rm tr} \Bigg[e \left[i\delta_{\mu 0} + \frac{\not{d}^i(q)}{2m} \delta_{\mu i} \right] \left[\frac{-i(q_0+k_0) + \not{d}(q)/(2m)}{(q_0+k_0)^2 + d(q)^2/(4m^2)} \right] \\
& \times e \left[i\delta_{\nu 0} + \frac{\not{d}^i(q)}{2m} \delta_{\nu i} \right] \left(\delta_{\mu 0}\delta_{\nu 0} \frac{v^2 q_\perp^2}{q_0^2 + v^2 q_\perp^4} + \delta_{\mu i}\delta_{\nu j} \frac{\delta_{ij}}{q_0^2 + v^2 q_\perp^4} \right) \Bigg] - 4\delta_1 (ik_0), \\
= & -\frac{2ie^2(d - 8m^2v^2 + 2)}{4m^2} \int_q \frac{(q_0+k_0) q^2}{(q_0^2 + v^2 q^4)((q_0+k_0)^2 + \frac{q^4}{4m^2})} - 4\delta_1(ik_0)
\end{aligned}
\tag{53}
$$

D. The photon self-energy

Now we turn to the counter terms in the photon sector. It shall be convenient to introduce the transverse and longitudinal projectors associated with the photon propagator, defined respectively as,

$$P^T_{\mu\nu}(k) = \delta_{\mu\nu} - \frac{k_\mu k_\nu}{k^2},\ P^L_{\mu\nu}(k) = \frac{k_\mu k_\nu}{k^2}. \tag{54}$$

We shall also define similar but separate projectors that act on the spatial components alone,

$$p^T_{ij}(k) = \delta_{ij} - \frac{k_i k_j}{k^2_\perp},\ p^L_{ij}(k) = \frac{k_i k_j}{k^2_\perp}. \tag{55}$$

With the aide of these projectors the one-loop renormalized photon self-energy can be written down as,

$$\begin{aligned}\Sigma_A^{\mu\nu,1\mathrm{L}}(k_0,\vec{k}_\perp) = &-\int_q \mathrm{tr}\Bigg[e\left[i\delta_{\mu 0} + \frac{\delta_{\mu i}\not{d}^i(k+2q)}{2m}\right] \\ &\times \left[\frac{-i(q_0+k_0) + \frac{\not{d}(q+k)}{2m}}{(q_0+k_0)^2 + \frac{|\vec{q}_\perp+\vec{k}_\perp|^4}{4m^2}}\right] \\ &\times e\left[i\delta_{\nu 0} + \frac{\delta_{\nu i}\not{d}^i(k+2q)}{2m}\right]\left[\frac{-iq_0 + \frac{\not{d}(q)}{2m}}{q_0^2 + \frac{q_\perp^4}{4m^2}}\right]\Bigg] \\ &- \delta_3\left(k^2 P^T_{\mu\nu}(k) - k^2_\perp \delta_{\mu i}\delta_{\nu j} p^T_{ij}(k)\right) \\ &- \delta_4 \delta_{\mu i}\delta_{\nu j} v^2 k^4_\perp p^T_{ij}(k).\end{aligned} \tag{56}$$

The contribution from the quartic vertex once again drops out identically due to the properties of the Λ_a matrices. The computation also greatly simplifies if we evaluate the self-energy at zero external frequency, $k_0 = 0$.

To extract the counter term δ_4 we separate the spatial components of the photon self-energy and evaluate it at the fixed external frequency $k_0 = 0$. We obtain,

$$\begin{aligned}&\Sigma_A^{ij,1\mathrm{L}}(0,\vec{k}) \\ &= -\int_q \mathrm{tr}\left[e\left[\frac{\not{d}^i(k+2q)}{2m}\right]\left[\frac{-iq_0 + \frac{\not{d}(q+k)}{2m}}{q_0^2 + \frac{|\vec{q}+\vec{k}|^4}{4m^2}}\right]\right. \\ &\quad \left.\times e\left[\frac{\not{d}^j(k+2q)}{2m}\right]\left[\frac{-iq_0 + \frac{\not{d}(q)}{2m}}{q_0^2 + \frac{q_\perp^4}{4m^2}}\right]\right] - \delta_4 v^2 k^4_\perp p^T_{ij}(k),\end{aligned} \tag{57}$$

$$= -\frac{e^2}{4m^2}\int_q \frac{M_1^{ij}(\vec{k},q_0,\vec{q})}{\left(q_0^2 + \frac{|\vec{q}+\vec{k}|^4}{4m^2}\right)\left(q_0^2 + \frac{q_\perp^4}{4m^2}\right)} - \delta_4 v^2 k^4_\perp p^T_{ij}(k), \tag{58}$$

$$\tag{59}$$

where,

$$M_1^{ij}(\vec{k},q_0,\vec{q}) = \mathrm{tr}\left[\not{d}^i(k+2q)\left(-iq_0 + \frac{\not{d}(q+k)}{2m}\right)\not{d}^j(k+2q)\left(-iq_0 + \frac{\not{d}(q)}{2m}\right)\right], \tag{60}$$

is the trace product in the numerator. We compute the frequency integral exactly using residue theorem and obtain,

$$
\begin{aligned}
&\Sigma_A^{ij,1\mathrm{L}}(0,\vec{k}) \\
=&-\frac{e^2}{4m^2}\int_{\vec{q}}\left(-\frac{1}{2}\frac{M_1^{ij}(\vec{k},i|\vec{k}+\vec{q}|^2/(2m),\vec{q})}{\frac{|\vec{k}+\vec{q}|^2}{2m}\left(\frac{|\vec{k}+\vec{q}|^2}{2m}-\frac{q_\perp^2}{2m}\right)\left(\frac{|\vec{k}+\vec{q}|^2}{2m}+\frac{q_\perp^2}{2m}\right)}+\frac{1}{2}\frac{M_1^{ij}(\vec{k},iq_\perp^2/(2m),\vec{q})}{\frac{q_\perp^2}{2m}\left(\frac{|\vec{k}+\vec{q}|^2}{2m}-\frac{q_\perp^2}{2m}\right)\left(\frac{|\vec{k}+\vec{q}|^2}{2m}+\frac{q_\perp^2}{2m}\right)}\right) \\
&-\delta_4 v^2 k_\perp^4 p_{ij}^T(k), \\
=&\frac{de^2}{4(d-1)^2 m}\int_{\vec{q}}\frac{M_2^{ij}(\vec{k},\vec{q})}{q_\perp^2\left(q_\perp^2+2\vec{q}\cdot\vec{k}+k_\perp^2\right)\left(q_\perp^2+\frac{2\vec{q}\cdot\vec{k}+k_\perp^2}{2}\right)} \\
=&\frac{2de^2}{4(d-1)^2 m}\int_{\vec{q}}\int_0^1 dx_1\int_0^1 dx_2\int_0^1 dx_3\frac{\delta\left(1-(x_1+x_2+x_3)\right)M_2^{ij}(\vec{k},\vec{q})}{\left(x_1 q_\perp^2+x_2\left(q_\perp^2+2\vec{q}\cdot\vec{k}+k_\perp^2\right)+x_3\left(q_\perp^2+\frac{2\vec{q}\cdot\vec{k}+k_\perp^2}{2}\right)\right)^3}, \\
=&\frac{2de^2}{4(d-1)^2 m}\int_{\vec{q}}\int_0^1 dx_1\int_0^1 dx_2\int_0^1 dx_3 \\
&\times\frac{\delta\left(1-(x_1+x_2+x_3)\right)M_2^{ij}(\vec{k},\vec{q})}{\left(q_\perp^2+2\vec{q}\cdot\left(x_2\vec{k}+(x_3/2)\vec{k}\right)+\left|x_2\vec{k}+(x_3/2)\vec{k}\right|^2+(x_2+x_3/2)\left(1-(x_2+x_3/2)\right)k_\perp^2\right)^3}, \\
=&\frac{2de^2}{4(d-1)^2 m}\int_{\vec{q}}\int_0^1 dx_1\int_0^1 dx_2\int_0^1 dx_3\frac{\delta\left(1-(x_1+x_2+x_3)\right)M_2^{ij}(\vec{k},\vec{q}-(x_2+x_3/2)\vec{k})}{\left(q_\perp^2+(x_2+x_3/2)\left(1-(x_2+x_3/2)\right)k_\perp^2\right)^3},
\end{aligned}
\tag{61}
$$

where in the last two lines we have used the delta function condition $x_1+x_2+x_3=1$ and shifted the internal momenta to simplify the integral.

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